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PUBLICLY AVAILABLE BUT NOT FULLY ACCESSIBLE: STRATEGIES FOR EQUITABLY SHARING POST-EARTHQUAKE LOSS INFORMATION

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***Abstract**: After an earthquake, the United States Geological Survey (USGS) produces near-real-time information on earthquake impacts through the publicly available PAGER (Prompt Assessment of Global Earthquakes for Response) loss estimation system. While PAGER was primarily designed for a targeted group of users—emergency responders, government and aid agencies, and the media—in reality, it reaches a much larger audience with tens of thousands of online viewers. In this study, we sought to understand how accessible and useful PAGER is to users with different backgrounds than the initial targeted user groups. This research was motivated by a concern that additional potential user groups were not being reached nor were they receiving the information that would be most useful to them. We draw on 16 focus groups with 49 disaster risk reduction practitioners who work with socially vulnerable groups in addition to USGS web traffic, and email lists. Our analysis of web traffic and email lists reveals that PAGER is widely accessed by users outside of earthquake-affected areas and in sectors outside of the user groups for which PAGER was initially designed. Our analysis of the focus group data indicated that PAGER could be more accessible to additional key user groups by making the information relatable, transparent, understandable, and shareable. Importantly, we find there are limits to the usability of the currently available PAGER information, where community-based organizations found it difficult to see how PAGER information could be helpful given their extensive on-the-ground knowledge. Considering these results, we offer strategies for developing post-earthquake loss information that is more accessible to a wider range of users while respecting the limits of a rapid loss estimation system such as PAGER.*

1 Introduction

After an earthquake, the United States Geological Survey (USGS) produces near-real-time information on the potential societal impacts of that earthquake through the PAGER loss estimation system. Specifically, PAGER (Prompt Assessment of Global Earthquakes for Response) (Figure 1) provides aggregate estimates of fatalities and economic losses with associated uncertainty (Wald et al., 2011) within about 25 minutes of an M3.4+ event in the United States, and M5.4+ worldwide (Earle et al., 2009). The PAGER system builds on other USGS near-real-time earthquake information tools including ShakeMap and “Did You Feel It?”, which are also made publicly available and aimed at both general and professional users.

PAGER expanded soon after the 2004 Indian Ocean Earthquake and Tsunami and became fully operational in 2007 (Jaiswal and Wald, 2011; Wald, 2023). In contrast to other USGS near-real-time products, like the ShakeMap, the PAGER loss estimation system was originally designed for professional users, or users whose job responsibilities require disaster response. It was funded and driven primarily by the U.S. Agency for International Development (USAID) and guided by multiple informal feedback sessions with other professional users in emergency management, like the Federal Emergency Management Authority (FEMA) (Wald, 2023). Since its initial release, PAGER has been produced online for over 6,000 events around the world obtaining

tens of thousands of online viewers for each significant event. While PAGER was initially developed for a targeted group of professional users like USAID and FEMA, its usage has grown and now informs additional organizations and even shapes public media discourse (Bialik, 2015).

PAGER is a breakthrough technical product that provides rapid, public, and validated earthquake impact estimates at a global scale. It is challenging, however, to maintain a loss estimation product, or any product for that matter, that is publicly available and understandable to such a wide and growing set of users. Many of PAGER's current users have variable levels of background or experience with earthquakes and/or loss information. Research has shown that groups with variable technical knowledge due to differing education, training, or personal experiences have differing abilities to understand disaster impact and forecast information (Smith et al., 2023). These differences lead to information inequities in that not all users have equal capabilities to access, understand, and, therefore, use PAGER information.

In this study, we sought to understand PAGER's accessibility for a broader range of user groups than the product's initial intended users. Here, we use a broad definition of accessibility that refers to "the extent to which products are able to be used by a population with the widest range of characteristics and capabilities to achieve a specified goal in a specific context" (page 524, Persson et al., 2015). We draw on several sources of primary data (focus group interviews) and secondary data (web analytics and email lists). From our analysis, we characterize PAGER's current users and identify strategies for improving PAGER's accessibility to a broader range of users. These strategies encompass more than improved visualization techniques (Karjack et al., 2022; Marti et al., 2023), but also focus on improving how PAGER is accessed, communicated, and ultimately used. The results of this research are already being used to influence how the USGS is sharing PAGER information; it can also inform other agencies involved in earthquake and other severe hazard impact forecasting efforts.

1.1 Rights to accessible information after a disaster

It is imperative for earthquake impact information, like PAGER, to be accessible to a broad range of stakeholders beyond its original design. In the influential report, *The Signal Code*, Greenwood et al. argue that access to information during a crisis is a basic humanitarian need, equally as important as food, water, shelter, and other humanitarian supplies (Greenwood et al., 2017). Further, they contend that information is increasingly important for informing emergency response decisions and enabling affected communities to become agents of their own protection. We note the definition of accessibility we employ here is a widened, yet complementary, definition to those that refer specifically to people with disabilities to consider a broader group of users who might not have previously accessed a product.

1.2 Previous studies of user-centered design and accessibility of post-disaster information

A means of achieving accessibility in the design of information systems is through user-centered design. User-centered design describes design processes that allow end-users to influence the look and function of systems or products (Abras et al., 2004). Human-centered design, a similar concept, prioritizes human needs through the development of a product, rather than a specific identified set of product users. Within design thinking methods, concepts of inclusive design and universal design are all approaches to support greater accessibility of information (Persson et al., 2015).

User-centered design in natural hazards research is often focused on usability testing to improve visualizations and/or the user interface/user experience (Cooper et al., 2022; Karjack et al., 2022; Luke et al., 2018; Marti et al., 2023; Radianti et al., 2019; Schneider et al., 2023; Thompson et al., 2015; Twomlow et al., 2022). Though more recently concepts of design have been expanded to other areas of emergency management (van Manen et al., 2023). Here, our approach draws from these methods but elucidates a larger set of strategies—visualization being among them—on the communication of PAGER information to be more accessible to a wider set of earthquake professionals and ultimately more members of the public.

A few studies have evaluated user needs regarding PAGER, primarily focusing on usability and visualizations. These studies have revealed that common practices in data visualization do not facilitate user understanding (Karjack et al., 2022; Marti et al., 2023). Through a case study, Marti et al. (2023) found that a transdisciplinary, iterative design process was mutually beneficial in developing earthquake rapid impact assessments and scenarios, allowing users to learn about products and incorporate them in planning while providing product developers with feedback early on to create more understandable products. They found that histograms of

probability that are standard practice were often misunderstood or misinterpreted. In a separate study evaluating PAGER's usability with undergraduate students, similar findings showed that users with non-technical backgrounds have difficulty understanding the information visualized in PAGER, struggle to understand uncertainty, estimation, and the scientific process, and were challenged to identify the importance of the estimates to inform earthquake response (Karjack et al., 2022). These results, although limited, suggest that the PAGER is not easily accessible and actionable for non-technical users.

2 Methods

The study described here is part of a larger project on developing human-centered earthquake impact information at the USGS. Through this project, we undertook an iterative approach to participatory, user-centered design that focused on multiple aspects, including evaluating the informational content of PAGER (which is outside of this paper's scope) and how PAGER's content is shared to improve its accessibility.

To understand what updates for accessibility purposes might be necessary, we assessed the users and use cases of PAGER through three sources of data: (1) analytics of USGS earthquake webpages, (2) email lists for those signed up to receive PAGER alerts, and (3) focus groups with select users. Webpage analytics provide a broad characterization of overall users of PAGER, email list analytics provide details on which organizations receive direct access to PAGER, and focus group data provides deeper feedback and reasons behind why users may or may not use PAGER.

2.1 Web and email list analytics

The two main ways in which users access PAGER are through an event page dynamically created after each earthquake on earthquakes.usgs.gov or through a direct email of the PAGER PDF (Earle et al., 2009). We analyzed these two modes of access through analysis of the users who visit the event page (via data from Google Analytics and the General Services Administration Digital Analytics Program) and the organizations signed up on the PAGER alert email list.

Google Analytics data provides information on multiple metrics associated with page visits or users on a given webpage (identified by a unique URL). Because of this, we can access information on webpage visitors for any earthquake where there is sufficient analytics data available, which for the earthquakes.usgs.gov website is after 2018. Using analytics data, we can identify key characteristics that can influence the design of a more accessible PAGER website, including when users are visiting a page, where they are visiting from, and what devices they are using. In our analysis, we use the provided Google metric of "users" to quantify engagement because it is the best metric for identifying individual viewers.

Our analysis of Google Analytics data compares web traffic to the PAGER website with other USGS near-real-time earthquake products, which includes the ShakeMap, the Aftershock Forecast, and Did You Feel It?, amongst others. We can analyze the visits to unique URLs associated with each product to determine which product a user has viewed. We analyzed 40 earthquake events that occurred between 2018 and 2023, are M3.5 or above, and had considerable web interaction (Appendix Table 2). Fifteen of the earthquakes in this sample occurred in the United States or its territories and the remaining 25 occurred internationally. Eight of these events received PAGER red alerts, ten received orange, seven yellow, and fifteen green. For each of these events, we analyzed web traffic from the first week after each earthquake.

We specifically evaluate two metrics we defined using location information from the Google Analytics data: (1) affected users and (2) general users. These metrics provide an aggregated summary of whether users of a product are more likely to be earthquake-affected populations in the area or region directly impacted by the event or those located outside of the affected region. Based on available data, we define affected users are those located in the states (if within the US and its territories) or in the countries (if outside the US and its territories) that experienced some level of shaking as indicated by ShakeMap, recognizing that this definition has limitations when earthquakes occur in large countries and states where not all populations are not equally impacted. General users are all other users.

While web traffic analysis provides a characterization of overall users of PAGER, analysis of the PAGER email list provides a more focused view of the current target users of PAGER. Since 2007, the USGS near-real-time development team has maintained an email list of recipients who receive PAGER alerts and summaries. At the start of this research in November 2021, this list included 478 unique individuals and organizations–who

expressed interest in receiving PAGER information via direct email and/or text message whenever a significant earthquake of interest occurs. Because some of the subscribers represent organizational distribution lists (for example, the respective Watch Offices for The White House, USAID, and FEMA), the emails are distributed to a much larger number of users than the 478 subscribers imply. To be added to the original USGS email list, a user must 1) know that PAGER exists and 2) be in contact with someone from the USGS involved with the PAGER system; however, again, because many people receive these messages via distribution lists, they do not need to subscribe directly through the USGS. At the time of subscription, users can select which alerts they receive based on geographic region(s) of interest and earthquake thresholds. They can also indicate whether they wish to receive email and/or text messages. Analyzing the people and organizations on the email list allowed us to identify which users have historically been connected to the USGS in terms of receiving PAGER communications, while also identifying which types of groups or organizations might be missing.

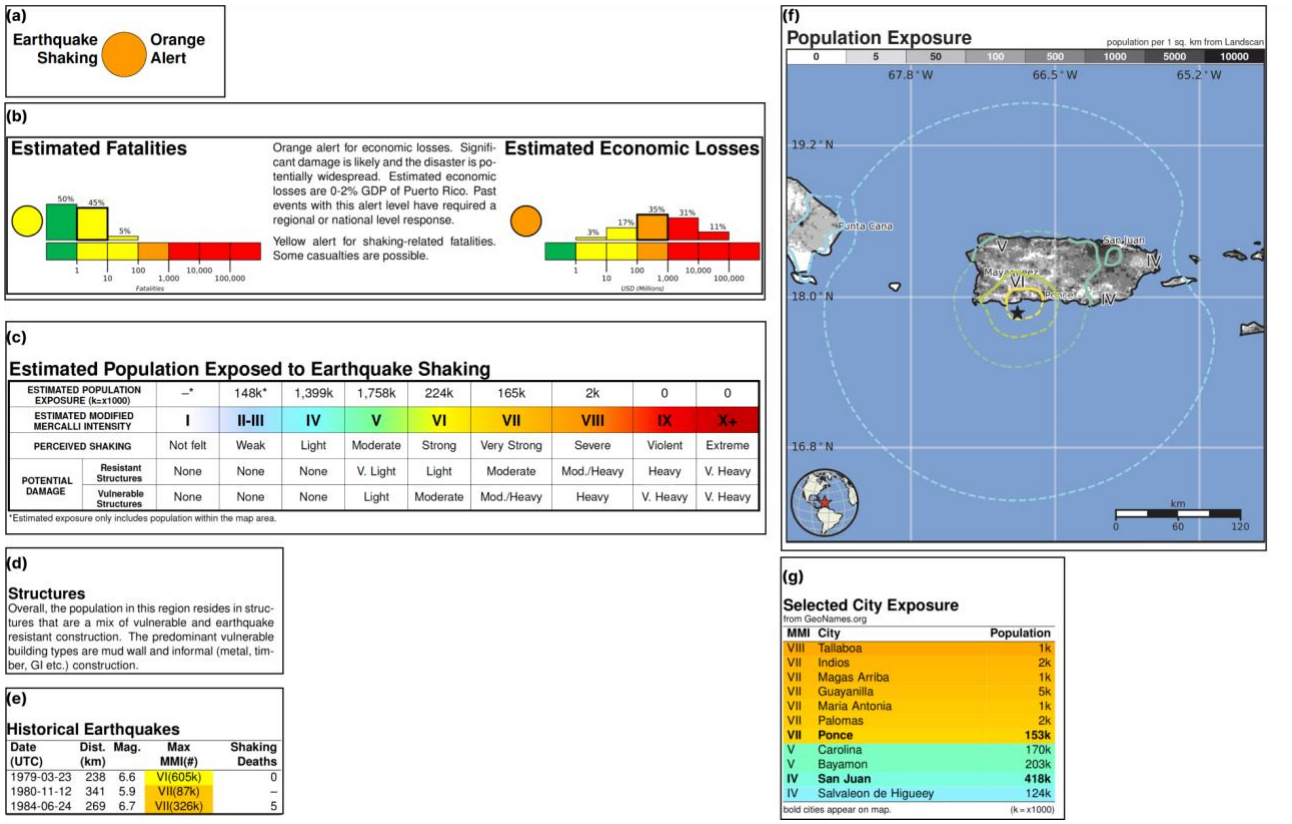


Figure 1. Components of the USGS PAGER from the 2020 Puerto Rico Earthquake used during focus groups

2.2 Focus groups

Between July and September of 2022, our team conducted 16 focus groups with 49 disaster risk reduction practitioners. Focus groups are small group interviews that provide a collaborative space for users to interact with each other and to share constructive feedback about a product or topic of research interest (Peek and Fothergill, 2009).

Webpage and email list analytics informed focus group design and participant selection. We purposively selected participants who are disaster risk reduction professionals representing a variety of sectors, including those identified in the PAGER email list. We selected participants who have varying levels of experience with PAGER but whose work prioritizes socially vulnerable groups exposed to earthquakes at multiple scales because their work often entails advancing equity and accessibility in earthquake risk reduction and response decisions. While our study population includes many disaster risk professionals, many of these professionals did not have frequent experience with earthquakes and the training or engineering risk that could be necessary to fully understand PAGER's content.

Each of the 90-minute virtual focus groups had between two to five participants and were conducted virtually using Microsoft Teams. One facilitator and one notetaker was present during each focus group. The facilitator would introduce the PAGER by providing an overarching summary of its availability and a more detailed summary of key elements of PAGER as shown in Figures 1(a) - (g). Participants shared how they do or likely would use PAGER information when making decisions during and immediately after an earthquake, with the ability to reference and provide use cases and feedback for each part of the PAGER.

Focus groups were recorded, transcribed, and then analyzed using a grounded theory approach, meaning instead of using predefined codes three members of the team read the data and then defined our codes inductively (Thornberg and Charmaz, 2014). This form of coding allowed the research team to reflect and synthesize results and quotes and how it relates to existing literature around accessible disaster information. The preliminary results were presented to the larger research team at a workshop in the fall of 2022, and then were further refined.

3 Current and key potential users of PAGER

3.1 Current users

The original design of PAGER was primarily to “more rapidly and effectively mobilize response” after an earthquake (page 2, Jaiswal and Wald, 2011), which aligns with common decision scenarios for originally intended PAGER users, including emergency managers and humanitarian aid workers from federal agencies such as USAID or FEMA. Often, these professionals are outside the affected area and are coming into the area to provide search and rescue operations or other relevant services as requested by local authorities.

Figure 2 confirms that the majority of direct recipients of PAGER are within these organizations. Here, we grouped organizations (submitted by the 478 email list members when they signed up for the email list) into sectors to preserve anonymity for smaller organizations. As evidenced by Figure 2, the top three sectors receiving direct PAGER alerts are within international development (45%), emergency management (18%), or geoscience agencies (15%). USAID recipients comprise 98% of the participants in the international development sector, FEMA comprises 72% of the emergency management recipients, and the USGS comprises 85% of the geoscience sector. In addition to these top three sectors, there are users representing the media, financial institutions, the military, humanitarian aid organizations, engineering firms, and beyond. As stated previously, these numbers does not capture everyone that receives email alerts given that some emails redistribute with other networks, but it does provide a sense of which organizations have most direct access to PAGER.

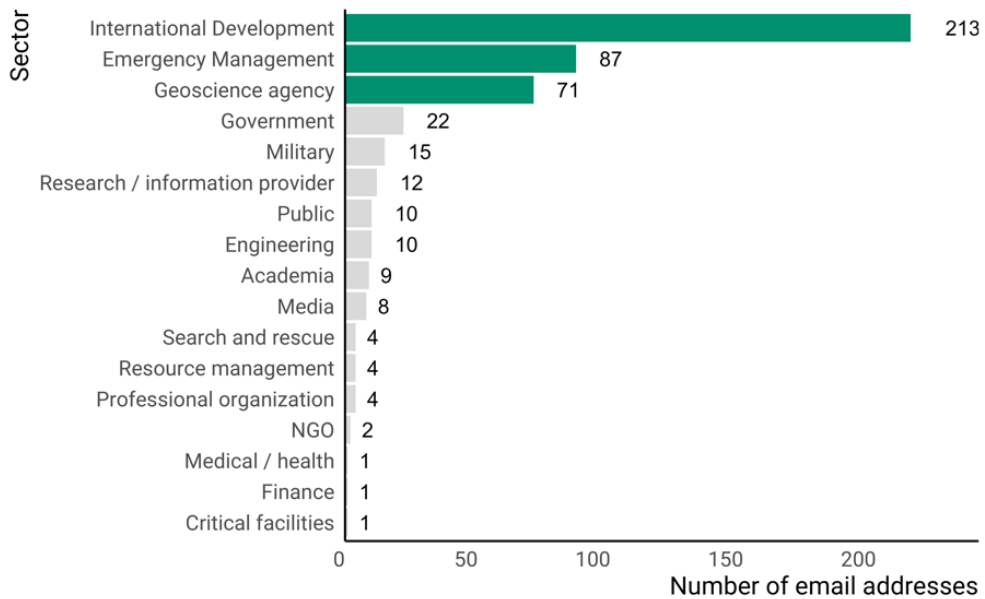


Figure 2. Summary of direct recipients of PAGER alerts from the PAGER email list.

However, several others have access to PAGER given its online availability, as evidenced by the analysis of webpage views. Figure 3 shows the number of users across all 17 USGS near-real-time products for all 40 earthquakes considered in the web traffic analysis (a) and only the red-alert events (b), split between affected and general users. As can be seen in Figure 3(a), PAGER is not the most visited product across all events, ranked fifth out of 17 products, but it is the most visited product when considering only red-alert events for both affected and general users. PAGER attracts substantial numbers of online users with events like the M7.2 Haiti earthquake attracting 19,000 users. Notably, 72% of PAGER's 190,700 total online visitors across these 40 events are from outside of the countries/states exposed to shaking by the earthquake. This aligns with our expectations given the design of PAGER to support the decision made by humanitarian or emergency responders to mobilize to an affected region.

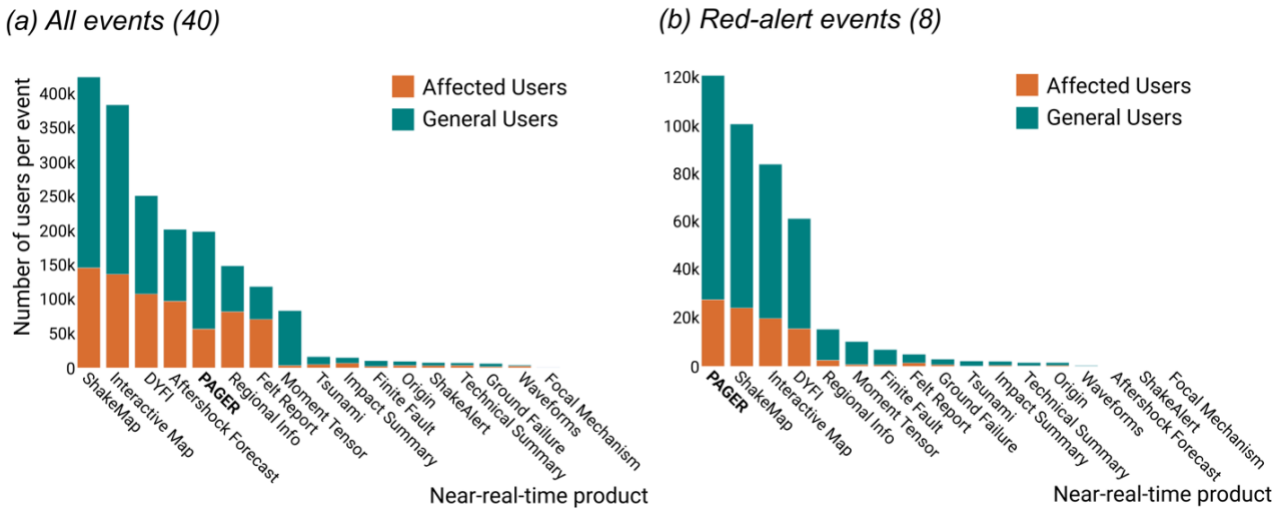


Figure 3. Summary of all users for (a) all 40 events and (b) only red-alert events.

From the web traffic data, we can also observe how online users access near-real-time products. The majority of users view near-real-time products from a mobile device rather than desktops or tablets.

3.2 Key potential users

Our focus group interviews included a range of participants from different organizations but also with varying degrees of familiarity with PAGER. Specifically, we actively selected participants who had both used PAGER and were highly familiar with the product, as well as those with limited or new familiarity. We refer to this latter group of users as "key potential users," which we define as those who could likely use PAGER more actively in their decision-making processes.

Through our focus groups, we were able to identify additional key potential users based on input provided by participants. Within emergency management, many state, county, and nongovernmental emergency managers were unfamiliar with PAGER. These users are important to consider as key potential users of PAGER since they would be immediately responding to future earthquakes at a more community-based level than federal emergency managers. These emergency managers are often responding to multiple types of disaster events and might have fewer experiences with earthquakes. It is therefore important to consider when designing PAGER for increased accessibility to an audience with less technical experience.

In addition, focus group participants who communicate with (e.g., media) or serve the public (e.g., policymakers and planners) viewed PAGER as a useful tool for communicating the impact of an event. These users might not have the same deep experience with earthquakes as more technical users, but more active inclusion could widen PAGER's user base to make it more accessible.

It is worth noting that most of the focus group participants worked outside of their own locally-earthquake-affected areas, though we did hold a focus group with community-based organizations involved in disaster response. We found in this focus group that the information in PAGER was not necessarily relevant for hyper-local organizations that will be on the ground after an event and would rely on more locally collected information.

4 Strategies to increase the accessibility of post-earthquake loss information

From our focus groups with current and key potential users, we have identified additional user needs for making PAGER more accessible. These needs center on four key areas defined below in Table 1. Each is foundational as to how the information connects to the PAGER user. Given the scope of this paper, we will focus on high-level summaries of selected needs.

Table 1. Summary of primary areas of user needs expressed in focus groups to make PAGER more accessible.

Area	Definition	Identified needs
Relatable information	Users can connect information with their experiences and understanding	• Interactive maps
Understandable information	Users can comprehend information	• Translations • Improved visualizations and iconography • Visual hierarchy
Transparent information	Users can identify background information necessary for critical use of the information	• Background on data sources and methods
Shareable information	Users can easily send or relate information to others	• Available in multiple formats (including raw data and low-bandwidth formats)

4.1 Developing relatable and understandable loss information

To develop relatable and understandable loss information, it is important to consider what aspects of the information enhance connection and comprehension of the disaster situation. From our focus group findings and claims backed by literature, we found that several components of PAGER are integral to users' ability to relate to and understand PAGER content, including the population exposure map (Figure 1(f)) and the estimated fatalities and estimated economic losses (Figure 1(b)). Key improvements users wanted to see to increase relatability and understandability for a broader range of users were an interactive map and language translation, respectively.

Regarding the relatability of PAGER, users frequently expressed that the population exposure map (Figure 1(f)) helped provide situational awareness, inform higher-up decision-makers, and share information with partners. Maps visually reveal information about affected areas, enhancing people's understanding of whether they or those they serve may have been personally affected (Dallo et al., 2022). This sentiment was most often expressed by emergency managers and information managers. There is still room for improvement in increasing the relatability of this map, especially regarding the format in which it is produced. Several users expressed the need for the ability to interact with the map (i.e., turn off and on shaking contours), especially citing the need for affected users to be able to connect the information on exposure and shaking to where they are located. The importance of an interactive map is evidenced by the popularity of the interactive map that already exists in the USGS near-real-time event page for any earthquake as shown in Figure 3(a). Currently, this map does not include final PAGER loss estimates due to uncertainties at this spatial resolution, but the static PAGER PDF could link users directly to the existing interactive map so they can interact with the online version of shaking and population exposure more easily.

Understandability of PAGER could be increased through visualization and translation. Currently, the histograms of the distribution of the estimated fatalities and estimated economic losses (Figure 1(b)) are used by many sectors for decision-making on local planning or distribution of funds; however, the way uncertainty is presented is not well understood. Several users expressed confusion and struggled to understand the both the meaning and the visualization of the histograms. Other studies have confirmed this issue with PAGER and have proposed alternative solutions (Karjack et al., 2022). As described by Wald (2023), the use of histograms is rooted in the PAGER team's goal to communicate the uncertainty of loss estimates to PAGER's technical users (Wald, 2023). Even within our focus groups, the most technical users who are familiar with loss

estimation methods expressed confusion about how to interpret this data, however. Given the repeated emphasis on the confusion of the histograms, it is clear that to make PAGER more accessible to a broader and larger user group, visualizations should be improved to more easily convey uncertainty and maintain the audience's understanding and trust.

Currently, PAGER is only produced in English even though the product is issued after global earthquakes of sufficient magnitude. Translation of disaster information is imperative to ensure that users globally have access to loss information communicated in PAGER. As explained by an information manager in one of our focus groups, *"It's easier for people who speak other languages to have an automatic translation of the document and it would be a very useful thing".* Globally, this could mean translating PAGER to other languages that would facilitate official response within the affected region. Within a country, language translation would enable those whose language is not the dominant language to be better agents of their own response. As described in *The Signal Code* disaster-affected populations have a right to information during crises (Greenwood et al., 2017) and language translation is a major tool to support communities in accessing this right (O'Brien et al., 2018).

4.2 Developing transparent and shareable loss information

Transparency is important to increase trust and reliability by revealing how information is produced and where the information comes from while holding information distributors accountable (Engebretsen et al., 2020). Further, transparency is a key strategy for responsibly deploying artificial intelligence technologies in disaster management (Soden et al., 2021) and arguably should also be for risk and loss models. Themes of transparency arose several times throughout the focus groups relating primarily to adding clearer descriptions of how each component of PAGER was developed and sharing the underlying data sources used to develop PAGER. Transparency about the underlying information in PAGER can empower PAGER users with more awareness for decision-making. As an information manager stated, "*I was also kind of confused about the probabilities because my first thought when looking at this is how to communicate the confidence in the data and where the data is coming from because I think that could really help with decisions."* As evidenced by this participant, it was unclear to several users where the information in PAGER was coming from, which then shaped whether and how they could use it to make decisions.

Sharing of PAGER data could allow users to access more information if they have the intention of doing more research on their own. For example, if downloadable GIS layers are shared, they then can be used in an approach suggested by a modeler where *"users that use the raw data and produce their own or that they can use this information to produce their own final reports."* In actuality, the USGS *does* share many of the results and underlying data sources to PAGER, like the ShakeMap, but several participants who are current users were not aware that these data were readily available.

Beyond being transparent about the underlying data and methods that contribute to the production of PAGER, many participants stressed the importance of producing PAGER in formats that could easily be shared amongst users who might not have access to a computer after an earthquake. Several available formats of PAGER information are important for users to be able to send during times of possible low bandwidth. Therefore, the emailed or texted PDF format is ideal for that need. The available information can then be shared internally within a response organization, externally with affected populations, and generally with the public on media platforms. The ability to share PAGER through mobile platforms is especially important as our web traffic analysis found that most PAGER users access it via a mobile phone. Developing easily sharable PAGER information is essential to ensuring everyone has access to relevant earthquake information during the time of the earthquake, acknowledging the availability of PAGER in several supports will support the overall communication of this product to several user groups or "publics" (Peek et al., 2016) Access to such post-disaster information can mitigate further loss of life, reduce emotional trauma, and direct people to essential services (Greenwood *et al.*, 2017).

5 Discussion and conclusions

This research was designed to characterize PAGER's current user base and to better understand how to reach additional key potential users to increase the accessibility of post-earthquake loss information. This study offers a broad understanding of PAGER's current users through email and web traffic analyses. We found that a large percentage of users that subscribe to PAGER emails belong in the originally intended user demographic

of humanitarian and emergency responders. We also argue, based on our analysis of the web analytics, that there is also broad interest in PAGER given its public availability, as evidenced by the tens of thousands of visitors to PAGER's webpage after recent major earthquakes. Through focus groups, we discovered four key areas of improvement to further increase PAGER's accessibility to key potential users as shown in Table 1. Specifically, our results show that the information in PAGER can be better presented and shared to facilitate users' abilities to *relate to, understand, understand,* and *share* its information.

Strategies to improve PAGER to facilitate these user abilities include improved visualizations as well as considering other aspects of how it is communicated with more diverse people and groups globally, including enhancing language translation, access to interactive formats, and including more background information. In many cases, the USGS has provided many of the resources to support these improvement strategies, but users who are unfamiliar with USGS products do not know where to access these resources. For example, the USGS near-real-time team developed a USGS PAGER Fact Sheet in 2010 that describes what is included and how each section of PAGER was developed (Wald, 2023; Wald et al., 2010), of which many users were not aware. Therefore, straight-forward additions to the current PAGER PDF would include better usage of dynamic hyperlinks to existing resources.

Based on this study and the larger project of which it is a part, the USGS is beginning to implement changes to PAGER communication. The existence of several sectors on PAGER's current email list (Figure 2) indicates interest in the product outside of humanitarian and emergency responders, however, the email list sign-up mechanism required subscribers to be in contact with someone from the USGS involved with the PAGER system. One tangible update the USGS is currently making includes adding PAGER to the USGS's Earthquake Notification Service (ENS, https://earthquake.usgs.gov/ens/) which is a free service that anyone can sign up for online to get alerts about global earthquake events. Previously ENS members were separate from PAGER email list members—thus ENS had an email list of 500,000 users whereas the PAGER email list was narrowed to a set of 478 targeted users. With the USGS adding PAGER information to ENS alerts, the overall accessibility of PAGER will be increased for many more users. Several other strategies are also being explored by PAGER leadership, including dynamic links, translation, and iconography.

Importantly, we find there are limits to who the currently available PAGER information is useful for—organizations that are embedded in their communities found it difficult to see how this information could be helpful beyond their already existing on-the-ground knowledge. This is not to say that post-earthquake loss information is not useful, however, it has limited usage at certain periods after an earthquake when other on-the-ground information is not readily available.

While this study was focused only on PAGER, the findings here may help inform other post-earthquake loss information systems including hurricane warnings and other extreme weather events (Soden et al., 2023; Taylor et al., 2018). For example, our methodology of combining secondary data analysis with qualitative insights can be helpful for informing other product improvements. More broadly, several ideas here can also expand existing user-centered design studies for information systems on other hazards to include accessibility and equity more directly in the design process. While it is unlikely that any hazard impact system will be a one-size-fits-all solution, it is possible to design these systems to be more accessible to a wider range of users through developing multiple accessible formats, relatable visualizations, and other strategies highlighted in this study. Improvements in these areas will lead to overall increased accessibility of PAGER facilitating the right to crisis information that all disaster-affected populations have a right to and need.

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8 Appendix

Table 2. Summary of 40 events considered in web traffic analysis

Year	Location	Name	Magnitude	Alert level
2018	Alaska	M 7.1 - Point MacKenzie, Alaska	7.1	Orange
2019	California	M 7.1 - Ridgecrest, California	7.1	Orange
2020	Massachusetts	M 3.6 - Bliss Corner, Massachusetts	3.6	Green
2020	Greece	M 7.0 - Néon Karlovásion, Greece	7	Orange
2020	Puerto Rico	M 6.4 - Maria Antonia, Puerto Rico	6.4	Orange
2021	Haiti	M 7.2 - Nippes, Haiti	7.2	Red
2021	India	M 6.0 - Dhekiajuli, India	6	Red
2022	Cyprus	M 6.6 - Pólis, Cyprus	6.6	Green
2022	Japan	M 7.3 - Namie, Japan	7.3	Red
2022	California	M 4.0 - Palomar Observatory, CA	4	Green
2022	Mexico	M 7.6 - Aguililla, Mexico	7.6	Orange
2022	Papua New Guinea	M 7.6 - Kainantu, Papua New Guinea	7.6	Orange
2022	California	M 6.4 - Ferndale, CA	6.4	Orange
2022	Turkey	M 6.1 - Duzce, Turkey	6.1	Orange
2022	Indonesia	M 5.6 - Sukabumi, Indonesia	5.6	Orange
2022	Iran	M 5.6 - Khowy, Iran	5.6	Orange
2022	California	M 5.1 - Alum Rock, CA	5.1	Green
2022	Oregon	M 4.4 - Lacomb, Oregon	4.4	Green
2023	Philippines	M 7.6 - Mindanao, Philippines	7.6	Orange
2023	Nepal	M 5.7 - Dailekh, Nepal	5.7	Orange
2023	California	M 5.4 - Rio Dell, CA	5.4	Orange
2023	Ecuador	M 6.8 - Balao, Ecuador	6.8	Orange
2023	Turkey	M 7.8 - Turkey earthquake sequence	7.8	Red
2023	Morocco	M 6.8 - Al Haouz, Morocco	6.8	Red
2023	Afghanistan	M 6.3 - Herāt, Afghanistan	6.3	Orange
2023	China	M 5.9 - Linxia Chengguanzhen, China	5.9	Red
2023	China	M 5.4 - Dezhou, China	5.4	Red
2023	Afghanistan	M 6.3 - Zindah Jan, Afghanistan	6.3	Orange
2023	Colombia	M 5.6 - Restrepo, Colombia	5.6	Orange
2023	South Africa	M 4.9 - Gauteng, South Africa	4.9	Green
2023	Afghanistan	M 6.5 - Jurm, Afghanistan	6.5	Green
2023	Indonesia	M 7.1 - Gili Air, Indonesia	7.1	Green
2023	Turkey	M 6.3 - Uzunbag, Turkey	6.3	Red
2023	Romania	M 5.6 - Lelesti, Romania	5.6	Green
2023	California	M 4.2 - Malibu Beach, CA	4.2	Green
2023	California	M 5.1 - Ojai, CA	5.1	Green
2023	Washington	M 4.3 - Port Townsend, WA	4.3	Green
2023	California	M 4.2 - Palomar Observatory, CA	4.2	Green
2023	California	M 4.2 - Isleton, CA	4.2	Green
2023	Texas	M 5.2 - Coalson Draw, Texas	5.2	Green